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GLP-1 Comparison Chart

Semaglutide | Tirzepatide | Retatrutide | Liraglutide

Dosing protocols, weight loss data, and reconstitution reference



GLP-1 Peptide Overview

GLP-1 (glucagon-like peptide-1) receptor agonists mimic the action of the natural incretin hormone GLP-1, which regulates appetite, gastric emptying, and glucose metabolism. These peptides are among the most researched compounds in metabolic health.

Compound Comparison

Compound	Targets	Half-Life	Dosing	Max Dose
Semaglutide	GLP-1 only	~7 days	Weekly	2.4 mg/wk
Tirzepatide	GLP-1 + GIP	~5 days	Weekly	15 mg/wk
Retatrutide	GLP-1 + GIP + Glucagon	~6 days	Weekly	12 mg/wk
Liraglutide	GLP-1 only	~13 hours	Daily	3 mg/day



Tirzepatide (Mounjaro / Zepbound)

Tirzepatide is a dual GLP-1 / GIP receptor agonist. GIP enhances the insulin response and may improve fat metabolism. It is the most prescribed GLP-1 peptide worldwide, with clinical data showing superior weight loss to semaglutide alone.

Standard Dosing Schedule

Weeks	Dose	Note
1-4	2.5 mg/wk	Initiation - GI adjustment period
5-8	5 mg/wk	First therapeutic dose
9-12	7.5 mg/wk	Intermediate escalation
13-16	10 mg/wk	Most common maintenance dose
17-20	12.5 mg/wk	High-range therapeutic
21+	15 mg/wk	Maximum approved dose

Weight Loss by Dose (SURMOUNT-1, 72 weeks)

Dose	Avg Weight Loss	5%+ Loss	20%+ Loss
5 mg/wk	16.0% (35 lbs / 16 kg)	89.4%	32.6%
10 mg/wk	21.4% (49 lbs / 22 kg)	96.2%	56.7%
15 mg/wk	22.5% (52 lbs / 24 kg)	96.3%	62.9%
Placebo	2.4% (5 lbs / 2 kg)	34.5%	1.3%

Note: The largest efficacy jump occurs between 5 mg and 10 mg. The incremental gain from 10 mg to 15 mg is modest (~1%), while side effect exposure increases.

Reconstitution Notes

- Common vial sizes: 5 mg, 10 mg, 15 mg, 30 mg
- BAC water: 1-2 mL typical. Use less for higher concentration / smaller injection volume.
- Refrigerate after reconstitution. Use within 4 weeks.



Semaglutide (Ozempic / Wegovy)

Semaglutide is a single GLP-1 receptor agonist with a ~7-day half-life achieved through albumin binding. It is the most extensively studied GLP-1 compound.

Semaglutide for weight management uses a higher dose range than for glucose control, with the key clinical trial (STEP) showing ~14.9% mean weight loss at 2.4 mg/week.

Standard Dosing Schedule

Weeks	Dose	Note
1-4	0.25 mg/wk	Initiation - GI tolerance only
5-8	0.5 mg/wk	Low therapeutic dose
9-12	1.0 mg/wk	Intermediate dose
13-16	1.7 mg/wk	High-range therapeutic
17+	2.4 mg/wk	Maximum dose for weight management

Semaglutide must be titrated slowly. Skipping dose levels increases GI side effects (nausea, vomiting). Each dose step should be maintained at least 4 weeks.

Weight Loss by Dose (STEP trials)

Dose	Avg Weight Loss
0.5 mg/wk	~8% at 68 weeks
1.0 mg/wk	~11% at 68 weeks
1.7 mg/wk	~13% at 68 weeks
2.4 mg/wk	~14.9% at 68 weeks

Reconstitution Notes

- Common vial sizes: 2 mg, 5 mg (multi-dose)
- BAC water: 1-2 mL. Low doses mean very small injection volumes -- use a 0.3 mL syringe for accuracy.
- Refrigerate after reconstitution. Use within 4 weeks.



Retatrutide

Retatrutide is a triple receptor agonist targeting GLP-1, GIP, and glucagon receptors. The added glucagon pathway increases energy expenditure and lipolysis. Currently in Phase 3 trials (expected approval late 2026-2027). Not yet FDA-approved.

Retatrutide has shown the highest weight loss of any incretin-based peptide in clinical trials, with the 12 mg dose producing up to 28.7% weight loss at 68 weeks.

Standard Dosing Schedule (Phase 2 / Expected Commercial)

Weeks	Dose	Note
1-4	2 mg/wk	Initiation
5-8	4 mg/wk	First escalation
9-12	8 mg/wk	Primary therapeutic dose
13+	Up to 12 mg/wk	Maximum dose

Weight Loss by Dose (Phase 2, 48 weeks)

Dose	24 Weeks	48 Weeks
4 mg/wk	~9.5%	~12.3%
8 mg/wk	~14.2%	~20.1%
12 mg/wk	~17.5%	~24.2%
Placebo	~1.6%	~2.1%

Additional data from Phase 3 TRIUMPH-4 showed 28.7% weight loss at 68 weeks on the 12 mg dose. Retatrutide also reduced liver fat by 86% at 12 mg -- a significant finding for metabolic dysfunction-associated steatotic liver disease (MASLD).

- **Important:** There is no direct dose conversion between tirzepatide and retatrutide. They are different compounds with different mechanisms.

Reconstitution Notes

- Common vial sizes: 5 mg, 10 mg, 15 mg
- BAC water: 1-2 mL. Higher concentrations needed for larger doses (8-12 mg).
- Refrigerate after reconstitution. Use within 4 weeks.



Liraglutide (Saxenda / Victoza)

Liraglutide is a once-daily GLP-1 agonist with a ~13-hour half-life. It requires daily subcutaneous injection. While less potent than the weekly peptides on total weight loss, it offers finer dose control for tolerability studies.

Standard Dosing Schedule

Weeks	Dose	Note
1	0.6 mg/day	Initiation
2	1.2 mg/day	First escalation
3	1.8 mg/day	Intermediate
4	2.4 mg/day	High-range
5+	3.0 mg/day	Maximum dose

Weight loss at 3.0 mg/day in SCALE trial: ~8% at 56 weeks.

Reconstitution Notes

- Often supplied as pre-filled pens at 6 mg/mL. Also available in 5-10 mg vials for research.
- Daily dosing means a 5 mg vial lasts 5-8 days depending on dose.
- Refrigerate. Discard 30 days after first use.



Head-to-Head Comparison

Dosing

Compound	Start Dose	Maintenance	Max Dose	Titration Steps
Semaglutide	0.25 mg/wk	1.0-2.4 mg/wk	2.4 mg/wk	4
Tirzepatide	2.5 mg/wk	5-10 mg/wk	15 mg/wk	5
Retatrutide	2 mg/wk	4-8 mg/wk	12 mg/wk	3
Liraglutide	0.6 mg/day	1.8-2.4 mg/day	3 mg/day	4

Weight Loss Results (Max Dose)

Compound	Trial	Weight Loss	Duration
Semaglutide	STEP	~14.9%	68 weeks
Tirzepatide	SURMOUNT-1	~22.5%	72 weeks
Retatrutide	Phase 2	~24.2%	48 weeks
Retatrutide	TRIUMPH-4	~28.7%	68 weeks
Liraglutide	SCALE	~8%	56 weeks

Side Effect Profile (General)

- Most common across all GLP-1s: nausea, vomiting, diarrhoea, constipation -- typically dose-dependent and worst during titration.
- Retatrutide has highest GI side effect rate due to triple agonism and greater potency.
- Liraglutide daily dosing can cause more injection site reactions.
- Semaglutide and tirzepatide have well-established safety profiles from large clinical trials.



FAQ

Which GLP-1 peptide is the most effective for weight loss?

Based on clinical trial data, retatrutide shows the highest weight loss (~28.7% at 68 weeks), followed by tirzepatide (~22.5%), semaglutide (~14.9%), and liraglutide (~8%). However, retatrutide is not yet FDA-approved and has a higher side effect profile.

Can I switch between GLP-1 peptides?

Switching protocols should be done carefully. There is no direct dose conversion between compounds. A common approach is to drop to the lowest initiation dose of the new compound and re-titrate. Always allow a washout period (at least 1 week for weekly peptides).

What's the difference between dual and triple agonism?

Dual agonists (tirzepatide) target GLP-1 and GIP receptors. Triple agonists (retatrutide) add glucagon receptor activation, which increases energy expenditure, thermogenesis, and lipolysis -- potentially driving greater weight loss and liver fat reduction.

Why do GLP-1s need gradual titration?

Rapid dose escalation causes significant GI side effects (nausea, vomiting, diarrhoea). The 4-week intervals at each dose step allow the body to adapt to increased receptor activation. Skipping steps does not accelerate results -- it increases dropout rates.

How long does it take to see results?

Some appetite suppression occurs within the first week. Measurable weight loss typically begins by week 4-8. Peak results are seen at 6-12 months depending on the compound and dose.

Can I reconstitute all GLP-1s the same way?

Yes. All lyophilised GLP-1 peptides follow the same reconstitution protocol: add BAC water, swirl gently, refrigerate. The key difference is dosing volume -- higher dose peptides (tirzepatide 15 mg, retatrutide 12 mg) may need more concentrated solutions or larger syringes. Use the ViralPeps Dosage Calculator to plan your reconstitution.

This chart is for educational and research reference purposes only. It does not constitute medical advice, diagnosis, or treatment recommendations. All peptides are for in-vitro research use only.

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